# vllm-sr/vela3-turbo

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## Name it. Vela finds it.

vLLM Semantic Router × KR Labs · Vela 3.0 family · **vela3-turbo** · 27B · base `d3-turbo` · Apache-2.0

What it does · Quick start · Evaluation · The family · Collection

**vela3-turbo** is the 27B member of the Vela 3.0 family, built on d3-turbo. In one `system_one` call it answers routing and general decisions and finds the exact spans you name (entities, relations, personal

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Model tree for vllm-sr/vela3-tur...
Base model vllm-sr/d3-turbo
Adapter (1) this model

Collection including vllm-sr/vela...
Vela 3.0 Collection
Six open routin... • 7 items • U... • △ 7

data, unsupported claims, evidence), with probabilities and character offsets and no generated text.

| | |
|---|---|
| **Parameters** | 26.09B with vision encoder (base) + 5044 MB adapters and heads |
| **Base** | vllm-sr/d3-turbo **v1.0**, revision `da17b5d0c860f7e5be6e2afebe67228b4` , unchanged (the public d3-turbo main is now package stays on v1.0) |
| **Inputs** | Text or JSON, plus images and videos for d3 own question types |
| **Answers** | Choice · Yes / No · Score · Span |
| **Licence** | Apache-2.0 |

## Summary

vela3-turbo is the reference model of the Vela 3.0 family, on d3-turbo v1.0 (pinned). d3-turbo, frozen, answers every ordinary decision, bit for bit, and gated adapters add the spans you name, entities, relations and router signals. It has the family's top scores on classification and relations.

## Highlights

- **Zero-shot classification:** 81.4 on the 13-set GLiFormer classification suite, above every model the paper reports (GPT-5-mini: 79.8, Gemma-4-31B-IT: 79.69; reported, GLiFormer paper Tab 5).
- **Relations:** 35.9 end to end (GLiFormer Table 4 protocol, avg of 4), above every model

reported (best: Qwen3.5-9B, prompted, 25.7; reported, GLiFormer paper Tab 4); 51.7 on CoNLL04 with the head entity given (GLiFormer-base: 8.6, our run, same harness).

- **Zero-shot NER:** 64.6 on OOD-7 (CrossNER + MIT) from label names, above every GLiNER / GLiFormer number reported on it (best: gliner-multitask-large-v0.5, 62.76).
- **Unsupported claims:** 0.900 example-F1 on the 10,698-answer LettuceDetect v2 test set, above LettuceDetect v2 mmBERT-base: 0.869 (reported, its card) and Vela 2.0 9B: 0.885 (its card).
- **Router signals:** better than Vela 2.0 9B on 20 of the 24 test files of the public router signal suite, level on 4, worse on none (paired 95% CI).
- **Decisions, bit for bit:** every ordinary decision is answered by `d3-turbo@da17b5d0` itself, adapters off (561 of 561 identical in our check).




vela3-turbo (27B) against reported numbers. Accent = vela3-turbo; grey = other models (reported).

## At a glance

| | |
|---|---|
| **What it is** | A 26B decision-and-extraction model for LLM routing and guardrails. It runs vllm-sr/d3-turbo, the largest d3 decision model, pinned to d3-turbo v1.0 (revision da17b5d0; the public main is now v1.1), unchanged, and adds the exact spans you name: entities, relations, personal data, unsupported claims and evidence. |
| **How it was trained** | d3-turbo stays frozen, so every ordinary decision is d3-turbo's own answer, bit for bit. Vela adds small gated LoRA adapters on top, one per job: spans (unsupported claims, personal data, evidence), entities, relations and router signals. Each adapter switches on only for its own questions. The adapters were trained on vela3-flash's recipe and ported to d3-turbo: span, entity and router adapters re-thresholded, the relation adapter retrained. Router signals are trained supervised, as in Vela 2.0, on the public router datasets Vela 2.0 used (for example AEGIS 2.0, PolyGuardMix and Nemotron-Safety-Guard v3 for safety, Global-MMLU for domain, WildFeedback for feedback, DiffusionDB for modality, MASSIVE for intents); domain and feedback also on the public router signal suite's train split. Other training data: public NER, relation, hallucination, evidence and PII datasets, plus web and document text labelled by a large teacher model. No test split and no Decision Index item is trained on; some router-signal-suite test rows also occur in public training sources, so the router table also reports the rows without that overlap. |
| **What it is good at** | Zero-shot classification; zero-shot relations; decisions, bit for bit; router signals above Vela 2.0 9B on 7 of 8 signals of the public router signal suite. Pick it for the most accurate answers, as the family's reference. |

| | |
|---|---|
| **How we measure it** | Public test sets only, run through this package one request at a time with the shipped thresholds: CoNLL04, CoNLL04 · DocRED · FewRel · CrossRE (avg), OOD-7 (CrossNER + MIT), RAGTruth, Super-NI task 1568, the 13-set GLiFormer classification suite, the LettuceDetect v2 test set, the public router signal suite, the Decision Index 0.4.1. Other models' numbers are as their authors report them on the same test set and metric, or scored by us on the same rows where a table says so. |

## What it does

| Capability | Ask (one question of a `system_one` call) | Get |
|---|---|---|
| **Decisions** | `{"type": "choice", "instructions": "Which topic?", "criteria": {"billing": None, "tech": None}}` (also `noul`, `score`) | a probability per option, from d3-turbo itself |
| **Router signals** | a built-in router question id from `router_questions.json` | the signal's answer from the gated decision adapter |
| **Named entities** | `{"type": "span", "criteria": ["drug", "dosage"]}` | spans with label, offsets and score |

| Capability | Ask (one question of a `system_one` call) | Get |
|---|---|---|
| **Relations** | `{"type": "span", "instructions": "Which spans are the founder of Apple?", "criteria": {"founder": None}}` | the related spans |
| **Personal data** | `{"type": "span", "criteria": ["PERSON", "EMAIL_ADDRESS"]}` | spans of the 17 trained PII types, any subset |
| **Unsupported claims** | state `{"context": ..., "answer": ...}` and `{"type": "span", "criteria": {"unsupported": None}}` | the words of the answer the context does not support |
| **Evidence** | `{"type": "span", "criteria": ["evidence"]}` | the sentences that support the answer |
| **Boxes, tracks, images in questions** | d3-turbo's `box` / `track` questions, `assets` | answered by d3-turbo itself, adapters off (d3-turbo card) |

## Quick start

```
# pip install "transformers==5.17.0" torch t
from transformers import AutoModel

m = AutoModel.from_pretrained("vllm-sr/vela3

# Name any labels; the spans come back with
r = m.system_one(
    state="Geoffrey Hinton presented capsule
    questions={"ents": {"type": "span", "cri
# researcher "Geoffrey Hinton", algorithm "c

# Unsupported claims in an answer, against a
r = m.system_one(
    state={"context": "The Eiffel Tower is 3
           "answer": "The Eiffel Tower is 33
    questions={"halu": {"type": "span", "cri
# flags "1925 by Gustave Eiffel's son"; add

# PII, relations and evidence use the same c
r = m.system_one(
    state="Steve Jobs founded Apple in Cuper
    questions={"pii": {"type": "span", "crit
               "rel": {"type": "span", "inst
```

## Evaluation

All numbers on public test sets, measured through this package with `system_one`, one request at a time and the shipped thresholds. Other models' numbers are as reported by their authors (or scored in the same harness where marked). Full per-set tables: EVALUATION.md.

▼ **Named entities**

| Benchmark | vela3-turbo | Best other model (reported) | Vela 2.0 9B | Size-class best (>10B reported |
|---|---|---|---|---|
| OOD-7, label names only (default; F1) | 64.6 | B2NER-20B 72.1 (standardised label names; B2NER Tab 3) | 43.5 | B2NER-20B standardised label names 72.1 (B2NER Table 3) |
| NER, 26 datasets of the GLiFormer paper (label names; not all zero-shot) | 63.7 | – | – | – |

Label names are the only input. Training holds no CrossNER or MIT text, though many generic CrossNER type names occur as training labels.

### ▼ Relations

| Benchmark | vela3-turbo | Best other model (reported) | Vela 2.0 9B | Size-class best (>10B, reported) |
|---|---|---|---|---|
| CoNLL04, head entity given, label names (F1) | **51.7** | GLiFormer-base 8.6 (same harness, Vela 2.0 card) | 10.1 | – |
| End to end, GLiFormer Table 4 | **35.9** | Qwen3.5-9B, prompted | – | Gemma-4-31B-IT, |

| Benchmark | vela3-turbo | Best other model (reported) | Vela 2.0 9B | Size-class best (>10B, reported) |
|---|---|---|---|---|
| protocol, avg of 4 (strict F1) | | 25.7 (GLiFormer paper Tab 4) | | prompted 25.1 |
| End to end: CoNLL04 · DocRED · FewRel · CrossRE (CoNLL04 zero-shot, others in-domain) | 42.8 · 33.7 · 27.8 · 39.2 | GPT-5-mini 42.4 · GLiNER-Relex 31.3 · Qwen3.5-9B 71.6 · GLiNER-Relex 18.1 (GLiFormer paper Tab 4) | – | – |

CrossRE 39.2 (in-domain) is above every model reported in the GLiFormer paper (best: GLiNER-Relex 18.1).

▼ Evidence

| Benchmark | vela3-turbo | Best other model (reported) | Vela 2.0 9B | Size-class best (>10B, reported) |
|---|---|---|---|---|
| QASPER, word-F1 (zero-shot) | 39.3 | – | – | – |
| ACL-Verbatim, | 49.1 | ACL-Verbatim | 24.5 | – |

| Benchmark | vela3-turbo | Best other model (reported) | Vela 2.0 9B | Size-class best (>10B, reported) |
|---|---|---|---|---|
| word-F1 (in-domain) | | ModernBERT 53.4 (in-domain, evidence only; Vela 2.0 card) | | |

ACL-Verbatim is in-domain: its silver training split is in the training data.

▼ **Unsupported claims (hallucination spans)**

| Benchmark | vela3-turbo | Best other model (reported) | Vela 2.0 9B | Size clas bes (>10 report |
|---|---|---|---|---|
| LettuceDetect v2 test set, 10,698 answers (example-F1) | 0.900 | LettuceDetect v2 qwen-2b 0.921 (its card) | 0.885 | – |
| RAGTruth (example-F1) | 0.779 | HDM-2 (3B) 0.850 (HalluciNot Tab 2) | 0.774 | Llama-13B fir tuned 0.787 (repor RAGTr Tab 5) |

▼ **Zero-shot classification**

| Benchmark | vela3-turbo | Best other model (reported) | Vela 2.0 9B | Size-class best (>10B, reported) |
|---|---|---|---|---|
| GLiFormer suite, 13 sets, label names only (mean macro-F1) | **81.4** | GPT-5-mini 79.8 (GLiFormer paper Tab 5) | – | Gemma-4-31B-IT 79.7 |

These questions run on d3-turbo with every adapter off.

▼ **Router signals** (public router signal suite; above Vela 2.0 9B on 7 of 8)

| Signal (metric, test rows) | vela3-turbo | Vela 2.0 9B, same rows | Decision-1.0-Route-0.6B, same rows | Rows sharing no text with training: vela3-turbo / Vela 2.0 9B |
|---|---|---|---|---|
| Harmful request (AUC, 2,230) | 0.958 | 0.957 | 0.963 | 0.953 / 0.958 (1,254) |
| Jailbreak or prompt injection (AUC, 1,878) | **0.967** | 0.919 | 0.991 | **0.965** / 0.902 (944) |

| Signal (metric, test rows) | vela3-turbo | Vela 2.0 9B, same rows | Decision-1.0-Route-0.6B, same rows | Rows sharing no text with training: vela3-turbo / Vela 2.0 9B |
|---|---|---|---|---|
| Personal data in the request (AUC, 3,000) | **0.971** | 0.960 | 0.994 | **0.973** / 0.960 (2,802) |
| Answer not supported by its context (AUC, 3,000) | **0.783** | 0.765 | 0.791 | 0.727 / 0.744 (1,743) |
| Fact-check needed (AUC, 800) | **0.870** | 0.809 | 0.942 | **0.863** / 0.805 (690) |
| Output modality (text or image) (AUC, 3,000) | **0.993** | 0.933 | 0.999 | **0.993** / 0.930 (2,916) |
| Domain (subject area) (accuracy, 2,996) | **0.884** | 0.765 | 0.863 | **0.885** / 0.763 (2,916) |
| User feedback on the previous answer | **0.852** | 0.666 | 0.848 | **0.843** / 0.631 (2,295) |

| Signal (metric, test rows) | vela3-turbo | Vela 2.0 9B, same rows | Decision-1.0-Route-0.6B, same rows | Rows sharing no text with training: vela3-turbo / Vela 2.0 9B |
|---|---|---|---|---|
| (accuracy, 2,842) | | | | |

vllm-sr/router-signal-suite (public), the `test` file of each signal, scored with the dataset's own scoring code. Each signal is a router question this package was trained on (supervised, as in Vela 2.0), asked one request at a time with the shipped thresholds.

Vela 2.0 9B was run by us through its public release on the same rows. Decision-1.0-Route-0.6B is scored from the suite's published predictions; it was trained on the suite's own train split, so these test files are in-distribution for it.

Bold: the paired 95% bootstrap CI against Vela 2.0 excludes a tie. The last column drops the test rows that share text (exact or a 13-gram) with this model's training data. Safety and jailbreak are scored on the rows whose text the suite publishes (2,230 of 3,003 and 1,878 of 3,000).

Domain and feedback are also trained on the suite's own train split (with the public router data Vela 2.0 used; the suite's test and hold rows, and anything sharing an exact text or a 13-gram with them, are kept out), as Decision-1.0-Route is, so their test files are in-distribution for both. The domain

question has a 15th option, religion, which the suite maps to philosophy. Feedback has NO_FEEDBACK as a trained option.

▼ **Decisions** (general decisions are the base model's, bit for bit)

| **Decision Index 0.4.1 (2026-10-28)** | **Public index** | **Full score (board rank)** |
|---|---|---|
| Fastino GLiDE 1.2 (tied rank) | 71.99 | 71.52 (1) |
| Perplexity Decider v1.2 (27B) (tied rank) | 67.74 | 67.98 (8) |
| vLLM-SR d3-turbo v1.1 (27B) (the public d3-turbo main; not this package's base; tied rank) | 71.62 | 67.31 (15) |
| vLLM-SR d3 (27B) (tied rank) | 65.45 | 67.15 (16) |
| GLiClass v4 Edge (0.4B) (tied rank) | 47.49 | 31.21 (156) |
| OneJev1.6-0.8B (tied rank) | 48.76 | 29.58 (162) |
| StartLux-Decision v1.1 0.8B (tied rank) | 53.79 | 28.88 (167) |
| **General decisions = d3-turbo** (d3-turbo v1.0, revision da17b5d0: not on the 0.4.1 board) | – | – |

General decisions are the base model's own answers, bit for bit, so a board entry of the base would be this package's decision score. Full score (the board's ranking): 60% private same-skill tests and 40% private new-domain tasks; public index: the 37 public benchmarks, chance-corrected. All

values are board entries on the Decision Index 0.4.1 board (multimodalart/jev-decision-index, data generated 2026-10-28T09:02Z) unless noted; the rank is the position in the board's Full-score order, and tied rank marks entries the board shows as tied. d3-turbo v1.0, this package's base (revision `da17b5d0`), is not on the Decision Index 0.4.1 board; the public d3-turbo main is v1.1, which is on the board at 67.31 (rank 15, tied).

| Benchmark | vela3-turbo | Best other model (reported) | Vela 2.0 9B | Size-class best (>10B, reported |
|---|---|---|---|---|
| Super-NI task 1568, held out (accuracy) | 0.636 | – | 0.574 | – |
| MASSIVE, three unseen scenarios: intent / scenario (accuracy; 14,911 decontaminated rows) | 0.833 / 0.937 | – | 0.763 / 0.871 (full set) | – |

MASSIVE: three scenarios never used in our router training (iot, transport, cooking), 52 locales; accuracy on the 14,911 test utterances that match no training text of the model's lineage. The 342 excluded utterances occur verbatim in upstream training mixtures (tasksource MASSIVE / HWU64 rows). Vela 2.0 values are on the full 15,253, so these rows are not bolded. Super-NI 1568: 79 to 90 of its

162 ProPara passages share a 13-gram with other tasks' training rows.

▶ **Latency**

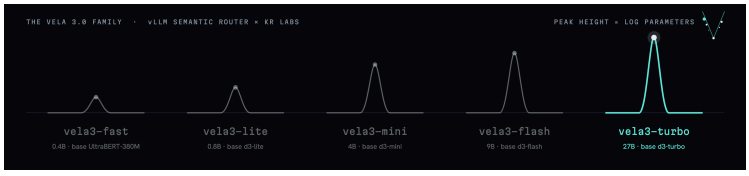
### Benchmarks

- **OOD-7** (CrossNER AI, literature, music, politics, science; MIT movie, restaurant): find the entities of the named types in text from domains the model was not trained on (strict entity F1).
- **NER, 26 datasets of the GLiFormer paper** (IE_INSTRUCTIONS): entities of named types across 26 public NER sets, not all zero-shot (strict entity F1).
- **CoNLL04, head entity given:** given one entity of a sentence, find the entities it is related to by the named relation (F1 over those tail entities).
- **Relations end to end, GLiFormer Table 4 protocol** (CoNLL04, DocRED, FewRel, CrossRE): find the entity pairs and their relation from raw text (strict triplet F1).
- **QASPER:** find the sentences of a research paper that answer a question about it (word-F1).
- **ACL-Verbatim** (KRLabsOrg/acl-verbatim-spans): find the passage of an ACL paper that supports an answer (word-F1).
- **LettuceDetect v2 test set** (10,698 answers, as in the LettuceDetect v2 paper (GroundLM 2026 workshop at EMNLP 2026) and card): does an answer contain claims its context does not

support (example-F1: the answer is flagged or not).

- **RAGTruth:** does a retrieval-augmented answer contain claims its retrieved context does not support (example-F1).
- **MASSIVE, three unseen scenarios** (iot, transport, cooking; 52 languages): pick the intent and the scenario of a user command the model never saw in training (accuracy).
- **Super-NI task 1568** (ProPara): answer a multiple-choice question about how a process changes an entity (accuracy).
- **Zero-shot classification, 13 sets** (the GLiFormer paper's suite: SST-2, IMDB, AG News, Banking77, and more): pick the label of a text from label names only (mean macro-F1).
- **Decision Index 0.4.1** (board, 2026-10-28): 37 public decision benchmarks plus private tests; the Full score ranks on the private same-skill and new-domain tests (chance-corrected skill, 0 = guessing).
- **Router signal suite** (vllm-sr/router-signal-suite): the vLLM Semantic Router's public test of its request signals (harmful requests, jailbreaks, personal data, unsupported answers, fact-check need, modality, domain, feedback); each label comes from public corpora (AUC or accuracy).

## The family



| Model | Size | Base | Best for |
|---|---|---|---|
| vela3-fast | 0.4B | UltraBERT-380M encoder | the default router; fastest: decisions + spans in one ~14 ms encoder pass, English, 32K |
| vela3-fast-vision | 0.3B | UltraVBERT encoder | image decisions, image-grounded hallucination spans, region pointing (preview) |
| vela3-lite | 0.8B | d3-lite | small GPUs, low latency |
| vela3-mini | 4B | d3-mini | balanced accuracy and speed |
| vela3-flash | 9B | d3-flash | high-accuracy routing |
| vela3-turbo | 27B | d3-turbo | the reference model |

## How it works

- **Frozen base.** `d3-turbo@da17b5d0` runs unchanged; ordinary questions are its own answers.
- **Gated LoRA adapters.** Each switches on only for its own questions, and every answer names the one it used (`head_reason`): decision (router (entity labels with definitions), relation, image (hallucination spans with images), span (evidence, PII, unsupported claims, toxic spans, names-only labels).
- **Spans as decisions.** The labels are presented as options; a span head scores every word

against every label, and agreeing neighbours form a span with character offsets.

- **Label names in.** Labels are read by name at this size; no definitions are written.
- **Specification.** Adds 5044 MB of adapters and heads; text or JSON state; requests with images, videos, assets, box or track questions, or long_context go to the base with every adapter off. Full evaluation: EVALUATION.md.

## Trade-offs

- **Unsupported claims:** hallucination spans 0.900 against 0.921 for LettuceDetect v2 qwen-2b, RAGTruth 0.779 against 0.850 for HDM-2 (3B) (reported).
- **NER from label names:** 64.6 on OOD-7, above the best reported encoder (gliner-multitask-v1.0 63.25) and below B2NER-20B, standardised label names 72.1 (reported).
- **End-to-end relations on generic types:** 35.9 average on the four typed sets, above every model in GLiFormer Table 4 (reported), but 0.2 on SciERC and 0.5 on NYT11, where every entity has the generic type `entity`.

## Citation

```
@misc{vela3_turbo,
  title        = {vela3-turbo (Vela 3.0, 27B
  author       = {{KR Labs} and {vLLM Semant
  year         = {2026},
  howpublished = {\url{https://huggingface.c
}
```